		Step Response			Frequency Response		
	Voltage gain at DC	Overshoot	Time to settle to 1%	Time to settle to 0.1%	Ripple in passband	Frequency for x100 attenuation	Frequency for x1000 attenuation
Bessel							
2 pole	1.27	0.4%	0.60	1.12	0%	12.74	40.4
4 pole	1.91	0.9%	0.66	1.20	0%	4.74	8.45
6 pole	2.87	0.7%	0.74	1.18	0%	3.65	5.43
8 pole	4.32	0.4%	0.80	1.16	0%	3.35	4.53
Butterworth							
2 pole	1.59	4.3%	1.06	1.66	0%	10.0	31.6
4 pole	2.58	10.9%	1.68	2.74	0%	3.17	5.62
6 pole	4.21	14.3%	2.74	3.92	0%	2.16	3.17
8 pole	6.84	16.4%	3.50	5.12	0%	1.78	2.38
Chebyshev							
2 pole	1.84	10.8%	1.10	1.62	6%	12.33	38.9
4 pole	4.21	18.2%	3.04	5.42	6%	2.59	4.47
6 pole	10.71	21.3%	5.86	10.4	6%	1.63	2.26
8 pole	28.58	23.0%	8.34	16.4	6%	1.34	1.66

TABLE 3-2
Characteristics of the three classic filters. The Bessel filter provides the best step response, making it the choice for time domain encoded signals. The Chebyshev and Butterworth filters are used to eliminate frequencies in the stopband, making them ideal for frequency domain encoded signals. Values in this table are in the units of *seconds* and *hertz*, for a one hertz cutoff frequency.

there are many ways for information to be encoded in an analog waveform, only two methods are common, **time domain encoding**, and **frequency domain encoding**. The difference between these two is critical in DSP, and will be a reoccurring theme throughout this book.

In *frequency domain encoding*, the information is contained in *sinusoidal waves* that combine to form the signal. Audio signals are an excellent example of this. When a person hears speech or music, the perceived sound depends on the frequencies present, and not on the particular *shape* of the waveform. This can be shown by passing an audio signal through a circuit that changes the phase of the various sinusoids, but retains their frequency and amplitude. The resulting signal *looks* completely different on an oscilloscope, but *sounds* identical. The pertinent information has been left intact, even though the waveform has been significantly altered. Since aliasing misplaces and overlaps frequency components, it directly destroys information encoded in the frequency domain. Consequently, digitization of these signals usually involves an antialias filter with a sharp cutoff, such as a Chebyshev, Elliptic, or Butterworth. What about the nasty step response of these filters? It doesn't matter; the encoded information isn't affected by this type of distortion.

In contrast, *time domain encoding* uses the *shape of the waveform* to store information. For example, physicians can monitor the electrical activity of a